Neurons:
Building Blocks of the Nervous System

You are driving down the road when suddenly your friend, who is sitting in the seat next to you, shouts: "Watch out for that truck!" Immediately, you experience strong anxiety, step on the brake, and look around in every direction. The process seems automatic, but think about it for a moment: How did information from your ears get "inside" and trigger your emotions and behavior? The answer involves the activity of **neurons**: cells within our bodies that are specialized for the tasks of receiving, moving, and processing information.

Neurons: Cells specialized for communicating information, the basic building blocks of the nervous system.

Dendrites: The parts of neurons that conduct action potentials toward the cell body.

Axon: The part of the neuron that conducts the action potential away from the cell body.

Glial Cells: Cells in the nervous system that surround, support, and protect neurons.

Axon Terminals: Structures at the end of axons that contain transmitter substances.

Synapse: A region where the axon of one neuron closely approaches other neurons or the cell membrane of other types of cells such as muscle cells.

Neurons: Their Basic Structure

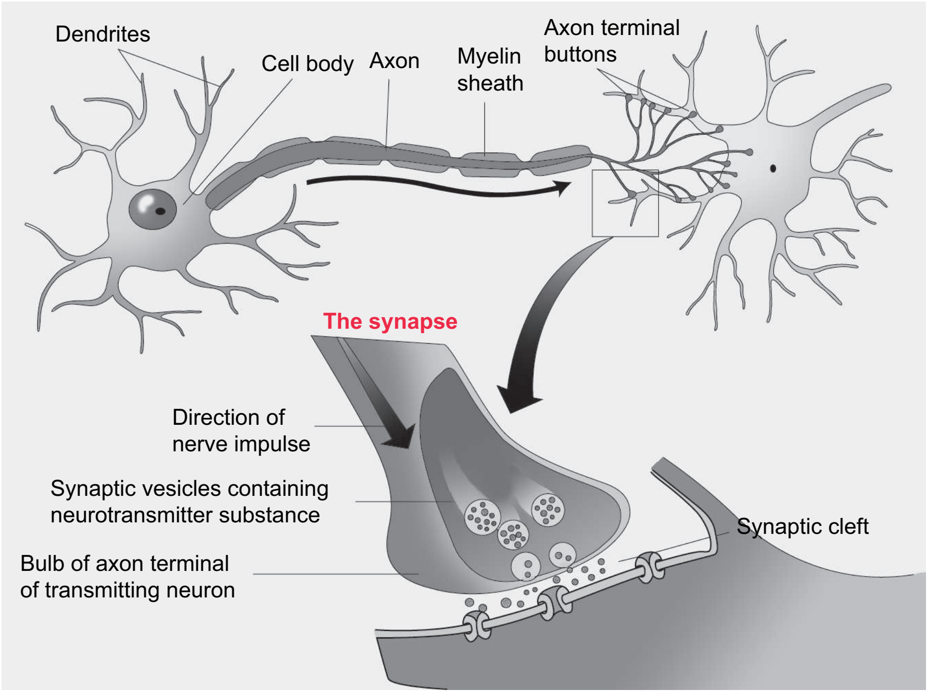
Neurons are tremendously varied in appearance. Yet most consist of three basic parts: (1) a *cell body*, (2) an *axon*, and (3) one or more *dendrites*. **Dendrites** carry information toward the cell body, whereas **axons** carry information away from it. Thus, in a sense, neurons are one-way channels of communication. Information usually moves from dendrites or the cell body toward the axon and then outward along this structure. A simplified diagram of a neuron and actual neurons are shown, magnified, in Figure 2.1. Scientists estimate that the human brain may contain more than 100 billion neurons.

In many neurons the axon is covered by a sheath of fatty material known as *myelin*. The myelin sheath (fatty wrapping) is interrupted by small gaps (places where it is absent). Both the sheath and the gaps in it play an important role in the neuron's ability to transmit information, a process we'll consider in detail shortly. Damage to the myelin sheath surrounding axons can seriously affect synaptic transmission. In diseases such as *multiple sclerosis* (MS), progressive deterioration of the myelin sheath leads to jerky, uncoordinated movement in the affected person.

The myelin sheath is actually produced by another basic set of building blocks within the nervous system, **glial cells**. Glial cells, which outnumber neurons by about ten to one, serve several functions in our nervous system; they form the myelin sheath around axons and perform basic housekeeping chores, such as cleaning up cellular debris. They also help form the *blood–brain barrier*—a barrier that prevents certain substances in the bloodstream from reaching the brain.

Near its end, the axon divides into several small branches. These, in turn, end in round structures known as **axon terminals** that closely approach, but do not actually touch other cells (other neurons, muscle cells, or gland cells). The region at which the axon terminals of a neuron closely approach other cells is known as the **synapse**. The manner in which neurons communicate with other cells across this tiny space is described next.

Figure 2.1
Neurons: Their Basic Structure
Neurons vary in form, but all possess the basic structures: a cell body, an axon (with axon terminals), and one or more dendrites.



Neurons: Their Basic Function

As we consider how neurons function, two questions arise: (1) How does information travel from point to point within a single neuron? And (2) how is information transmitted from one neuron to another or from neurons to other cells of the body?

Communication Within Neurons: Graded and Action Potentials

The answer to the first question is complex but can be summarized as follows: When a neuron is at rest, there is a tiny electrical charge (–70 millivolts) across the cell membrane. That is, the inside of the cell has a slight negative charge relative to the outside. This electrical charge is due to the fact that several types of ions (positively and negatively charged particles) exist in different concentrations outside and inside the cell. As a result, the interior of the cell membrane acquires a tiny negative charge relative to the outside. This resting potential does not occur by accident; the neuron works to maintain it by actively pumping positively charged ions back outside if they enter, while retaining negatively charged ions in greater concentrations than are present outside the cell.

Graded Potential: A basic type of signal within neurons that results from external physical stimulation of the dendrite or cell body. In contrast to the all-or-nothing nature of action potentials, graded potentials vary in proportion to the size of the stimulus that produced them.

Stimulation, either directly (by light, heat, or pressure) or by chemical messages from other neurons, produces **graded potentials**—a basic type of signal *within* neurons. An important feature of graded potentials is